

Fluid Flow in Pipe Systems

Final Lab Report

Unit Operations Lab 5

November 11, 2019

Group #2, Tuesday, PM section

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1. Executive Summary (10 pts)

The objective of this study is to compare experimentally acquired data from a fluid flow system, the Hampden H-6920 Pipe Friction Demonstrator, to their theoretical values. This data, used to calculate energy loss coefficients like friction factor (f), loss coefficient (K), and flow coefficient (C_f), are compared to theoretical values, such as values calculated from the Haaland equation or obtained from the Moody diagram. The parts of the fluid flow system being measured include an orifice meter, an expansion and contraction pipe, 90° short and square elbows, a 180° U-bend, and straight pipes made of plastic, copper, and stainless steel, all at different diameters.

The first step required a calibration of the rotameter system used to measure the flowrate of the water. This was done successfully by converting the units on the rotameter, standard cubic feet of air, to liters per minute of water. A linear fit of the data with a line equation of $y = 1.05x - 3.46$ and an R^2 value of 0.98 was used for all calibrated conversions.

The second part of this study included calculating the flow coefficient, C_f , at various flow rates across the orifice meter. Tabulated values range from 0.6 to 0.9 [1]. We obtained values ranging from 0.52 to 0.65, from lowest flowrate to highest. This indicates that C_f changes with flowrate.

Third, we looked at all fluid flowing through straight pipes made of either copper, plastic, or steel. Using the Haaland equation and Moody diagram [1,2], we compared experimentally obtained values f to empirical values. Graphical representation of f as a function of Reynold's number, Re , can be viewed in the Results section under "Pipe Friction Head Loss" on pages 8 and 9, graphs 4.3-4.5. In both empirical and experimental calculations, the friction factor shows a negative correlation with the Reynolds number. Experimental data and calculations can be found in Appendix I, Tables II-X. The errors in our experimental values could be explained by the leaking apparatus or perhaps the non-linearities found in our rotameter calibration.

Lastly, for all pipe fittings (except the orifice meter) the loss coefficient K was calculated and graphed as a function of Reynolds number. Graphs can be seen in the results section under "Pipe Fitting Head Loss" on page 10. What we found was that the loss coefficient decreases with an increasing Re , such as in the 180° U-bend, where the K value at $Re = 31966$ is 0.0087 and at $Re = 19487$ is 0.1439. We also obtained an equivalent length of diameter, L , for the fittings based on 50 pipe diameters, as used in ordinary practice [3]. This L value came to be 1.33 meters.

2. Introduction (10 pts)

The objective of this experiment was to obtain and analyze data to determine the friction factor f , the loss coefficient K , and the flow coefficient C_f in different parts of a pipe network – Hampden H-6920 Pipe Friction Demonstrator – as a function of water flowrate. The values obtained were then compared to empirical correlation values, which made it possible to differentiate the Fanning friction factor and the Darcy friction factor.

Prior to the experiment, a calibration curve was made by taking multiple water samples at different flowrates by observing the time with a stopwatch and the volume of water collected in the graduated cylinder. This procedure was done in order to calibrate the rotameter, since this piece of equipment displays a standard cubic foot per minute of air (SCFM) and not for water. After the conversion factor was determined, the experiment proceeded.

The set of measurements that followed the experiment was separated into two parts. The first part of this experiment consisted of measurements throughout the straight sections of the pipe. The pipe network was composed of three different materials, PVC, copper and stainless steel. Each of those materials was composed of three different pipe diameters. The water flowrate, the pressure drop in the pipe section, and the water temperature were the three measurements taken during this part of the experiment in order to determine the energy loss coefficient and the friction factor. In order to do so, the reservoir tank was filled with water, the valves were opened in the pipe system and the pump was turned on. Before the quick connection tubing was attached to the meter, the pressure transmitter was observed to give a reading of 0.00, which enable the continuity of the experiment. By starting from the lowest diameter, the pipe being measured was the only one with an open valve, insulating the rest of the system. The two quick connections were inserted in the form that the “upstream pressure sensing port is connected to the high side of the DPT and the downstream pressure sensing port to the low side of the DPT” [1] After the reading in the transmitter stabilized, all the data specified above was collected and the experiment proceed to another measurement. Varying water flowrates from 15 SCFM to 48 SCFM were used for each type of pipe and for all the three diameters available. The same water flowrates were used for the remaining sections of the experiment.

The second part of the measurements took place in the different fittings composed of 180° U-shape bend, a straight 90° bend, 90° bend, an orifice meter, a compression section and an expansion section, where the water flowrate, pressure drop throughout the pipe section being measured and the temperature were recorded.

All the pipes and all the fittings were measured three times with three different water flowrates, except for the orifice, which was measured five times, with five different water flowrates. This experiment makes possible to determine the friction factors experimentally as a function of Reynolds number, and the observation that the flow coefficient should increase with the volumetric water flowrate.

As an industrial application, the study of fluid flowing in pipe system is extremely important when designing a pipe network, as it is necessary to know the pressure drop through the system, which includes straight lines and fittings, in order to determine the pressure drop, the flow

coefficient and energy lost. All those variables must be taken into consideration when choosing the material of the pipe, its diameter, the flowrate applied and the conditions of the fluid.

3. Theory (10 pts)

In 1883 Osborne Reynolds demonstrated through his experiment the behavior of water flow under various flowrates, which the fluid transitioned from laminar flow to turbulent flow. Under his study, Reynolds determined the ratio of inertial forces to the viscous forces and gave it the name of Reynolds Number, an unitless measurement [4].

$$Re = \frac{Dv\rho}{\mu} \quad (1)$$

Where D is the diameter in meters, ρ is the fluid's density in $\frac{kg}{m^3}$, μ is the fluid's viscosity in $Pa \cdot s$, and v is the average velocity of the fluid in $\frac{m}{s}$. For a straight circular pipe, if Reynolds number is less than 2100, the flow is considered laminar flow. If the value is above 4000 the flow is considered turbulent flow [4]. Laminar flow can be solved analytically, while turbulent flow must rely on semi-empirical theories and data obtained experimentally [5]. The majority of the flows encountered in engineering are the turbulent type [6], and this will be the focus on this experiment.

This experiment makes use of an orifice meter, which is a type of flow meter that utilizes differential pressure measurement principles to determine the rate of fluid flowing through. When the device is placed in the pipeline, a pressure drop is generated between the inlet section and outlet section of the orifice meter, this pressure differential is linear and direct proportional to the flowrate of the fluid and can be described by the incompressible Bernoulli's equation [1]:

$$\Delta p = p_1 - p_2 = \frac{1}{2} \rho V_2^2 - \frac{1}{2} \rho V_1^2 \quad (2)$$

Where stream 1 is the upstream of the orifice, and stream 2 is the downstream. The continuity equation makes possible to calculate the pressure differential in terms of the cross-sectional area of the pipe and the volumetric flowrate of the fluid [1]:

$$\Delta p = \frac{1}{2} \rho Q^2 * \frac{1}{A_2^2} * \left[1 - \left(\frac{A_2}{A_1} \right)^2 \right] \quad (3)$$

In order to calculate the flowrate for the turbulent flow, a discharge coefficient C_d must be taken into consideration as the viscosity and turbulence converts part of the kinetic energy into heat, reducing the flowrate [1].

$$Q = C_d \sqrt{\frac{2\Delta p}{\rho}} * \frac{A_2}{\sqrt{1 - \left(\frac{A_2}{A_1} \right)^2}} \quad (4)$$

However, when calculating the flow in an orifice meter, things get a little bit more complicated. "The fluid stream, once past the orifice plate, forms a vena contracta or free-flowing jet" [4], which makes the value of A_2 uncertain. In order to minimize this flaw, a flow coefficient C_f is introduced [1].

$$C_f A_0 = C_d * \frac{A_2}{\sqrt{1 - \left(\frac{A_2}{A_1}\right)^2}} \quad (5)$$

Where A_0 is the orifice's area. Making possible to calculate the flowrate Q for real flows as:

$$Q = C_f A_0 \sqrt{\frac{2 * \Delta p}{\rho}} \quad (6)$$

The flow coefficient ranges from 0.6 to 0.9 and it is analytically determined, however it's possible to find references to it in the literature as β , which is a ratio of the diameter of the orifice to the pipe inlet diameter [1].

$$\beta = \frac{D_o}{D_{inlet}} \quad (7)$$

The mass flowrate can be calculated by multiplying the density of the fluid, by the flow observed. In this experiment, the rotameter was used as a measurement equipment of the flow. In this device, “the fluid enters at the bottom of the tapered glass tube, flows through the annular area around the tube, and exits at the top” [4]. During this process, the flow remains stationary, and the pressure differential remains constant. Since this device measures standard cubic feet per minute of air (SCFM), a conversion had to be done in order to obtain the value of the flowrate in liters per minute of water, and this relationship is approximately linear.

$$Q_{mass} = \rho * Q \quad (8)$$

The friction factor f , or resistance coefficient, for laminar flow depends only on Reynolds Number, however, for the turbulent flow f depends on Reynolds number and the roughness of the pipe and must be determined empirically. Since this measurement can't be determined theoretically for the turbulent flow in order to be calculated, the pressure drop must be known [1].

$$f = \frac{\Delta p D}{2L\rho V^2} \quad (9)$$

The roughness of the pipe can also be calculated as a function of equivalent sand roughness of the pipe by the pipe's diameter $\frac{K_s}{D}$ (10).

When trying to use a chart of f versus Modified Reynolds Number as **Figure 3.1** shows, it is good to pay attention to each friction factor being used in the graph, since the Darcy friction factor can be as large as four times the Fanning friction factor [4].

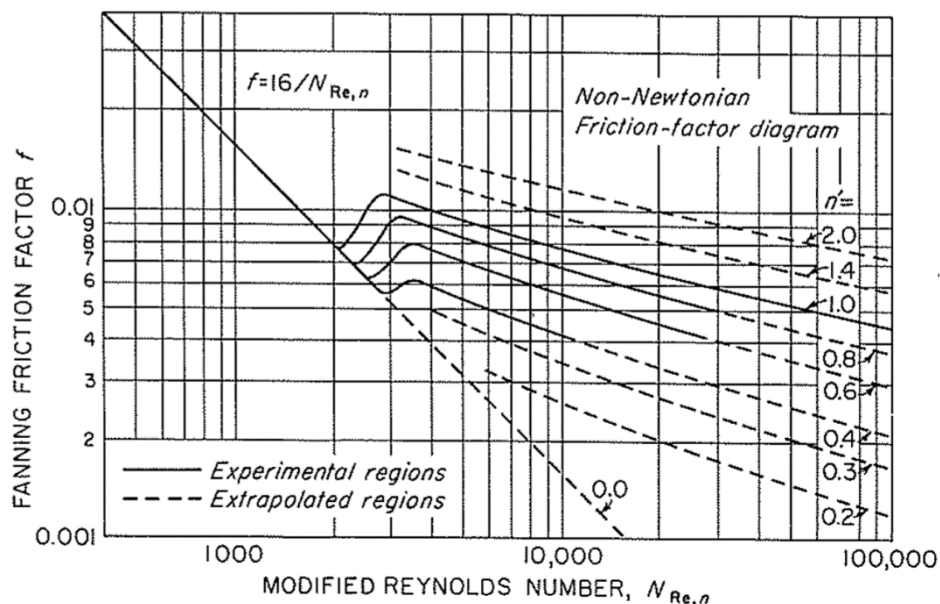


Figure 3.1: Fanning Friction Factor chart [1].

Heat loss due to friction is another parameter that must be analyzed, since the temperature gradient can cause changes in the physical properties of a fluid, which includes viscosity [4]. And so, the heat loss can be calculated if the friction factor of the system is known:

$$h_L = \frac{\Delta p}{\rho} = f \frac{L V^2}{D} \quad (11)$$

It is possible to calculate the friction loss in the fluid through a straight pipe by only using Fanning friction factor. However, in the case where there is a change in the fluid's velocity either in direction or magnitude, additional friction losses may be needed to be taken into consideration, such as pipe expansions or contractions and also for pipe's fittings and valves. "If the cross section of a pipe enlarges very gradually, few or no extra losses are incurred. If the change is sudden, it results in additional losses due to eddies formed by the jet expanding in the enlarged section" [6]. The same occurs for contractions, if the cross section is reduced suddenly, the fluid has difficulties flowing through the new sharp corners, and additional friction must be accounted for due to eddies occurrent. The pressure differential can be pointed as the primary cause of the energy loss. Due to a flow separation, additional turbulence may be generated, such disturbance can be observed in the form of a swirling, which will result in energy loss [1]. For pipe fittings, the head loss can be estimated by:

$$h_L = K \frac{V^2}{2} = f \frac{L * V^2}{2D} \quad (12)$$

Where K is the loss coefficient. Empirical data from commercial sources, can be used as an estimate of head loss. It is also common to determine the head loss as a function of the length of diameters, L, by taking in consideration the pressure differential throughout the whole system and by adding the equivalent length of the straight pipe section and all the fittings presented in the pipe network. By doing so, it is possible to determine the head loss for the whole system.

4. Results (15 pts)

Rotameter Calibration

To calibrate the conversion factor for the rotameter, which is marked in standard cubic feet per minute (SCFM) of air instead of liters per minute (LPM) of water, a series of six flow measurements were performed. Various SCFM flowrates were used. Based on the volume of water that flowed out in that time, LPM was obtained by dividing the volume of water by the time. A calibration curve of LPM vs. SCFM was used to calculate all flowrate measurements into LPM. The linear fit of this curve is $y = 1.05x - 3.46$ with an R^2 value of 0.98.

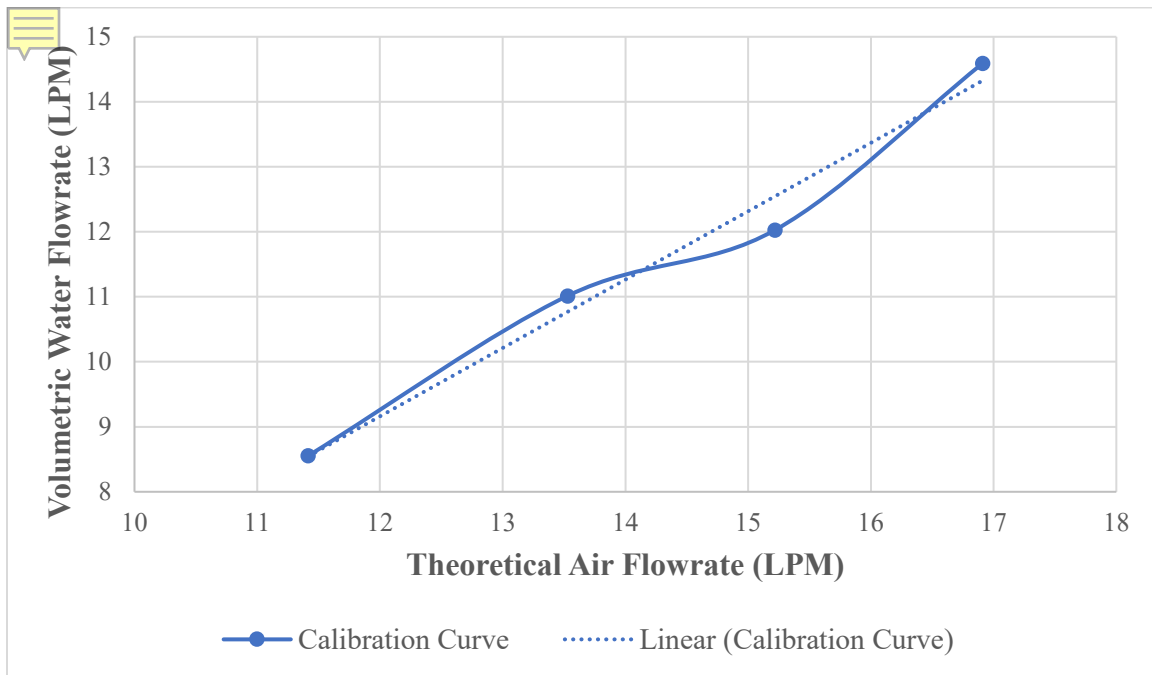


Figure 4.1: Calibration Curve of the Rotameter.

Orifice Meter

For the orifice coefficient, five different flow rates were used to obtain C_f . Values of C_f ranged from 0.52 to 0.65. Data for this section can be viewed in Appendix I, Table XVI. As Figure 4.2 below shows, C_f does indeed change with varying flow rates.

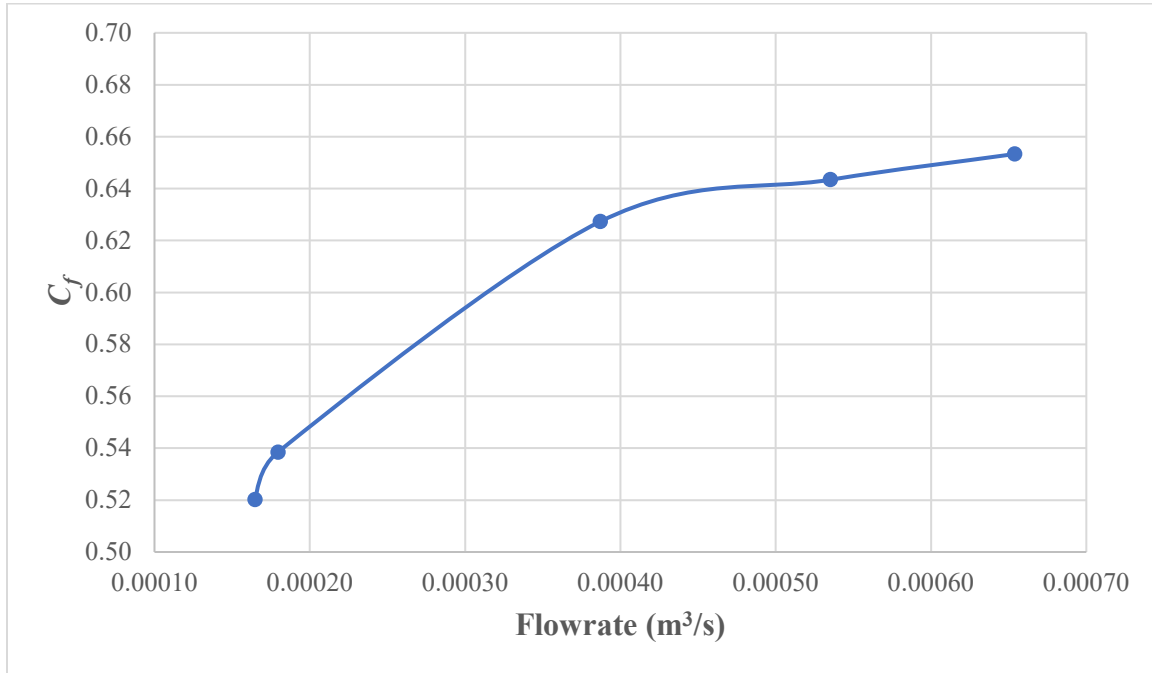


Figure 4.2: Orifice coefficient, C_f , vs. Flowrate.

Pipe Friction Head Loss

For each tube material and diameter of straight pipe, three differing flowrates were used to obtain the friction factor, f , experimentally. To compare experimental friction factors to empirical friction factors, we used the Haaland Equation [6]:

$$\frac{1}{\sqrt{f}} = -3.6 \log_{10} \left\{ \frac{6.9}{Re} + \frac{k}{3.707D} \right\} \quad (13)$$

All raw data and tabulations can be found in Appendix I, Tables II-X. Graphs of, f as a function of Reynold's number can be seen below in Figures 4.3-4.5.

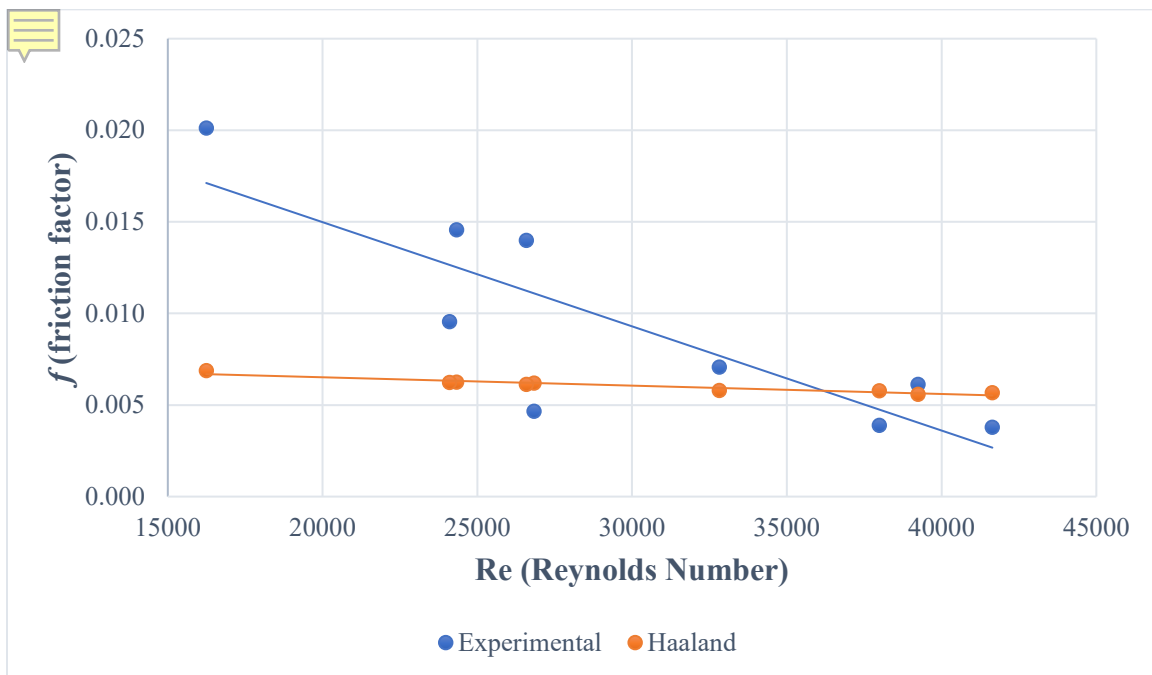


Figure 4.3: Friction factor, f , vs. Reynold's Number, Re , for Copper Pipe.

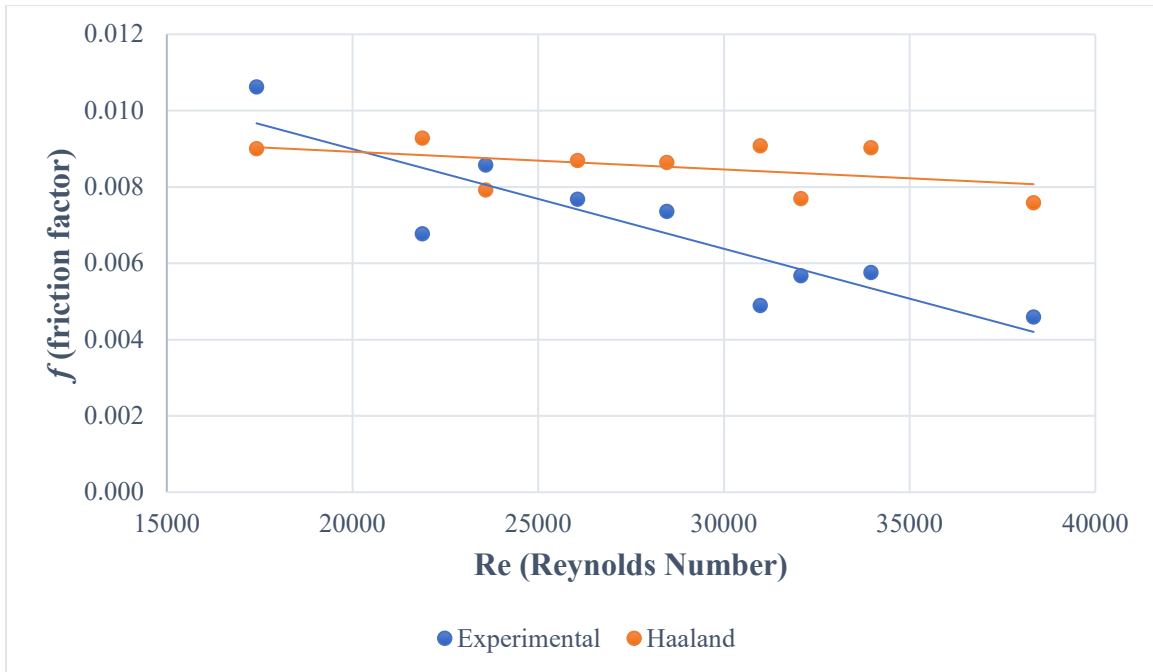


Figure 4.4: Friction factor, f , vs. Reynold's Number, Re , for Steel Pipe.

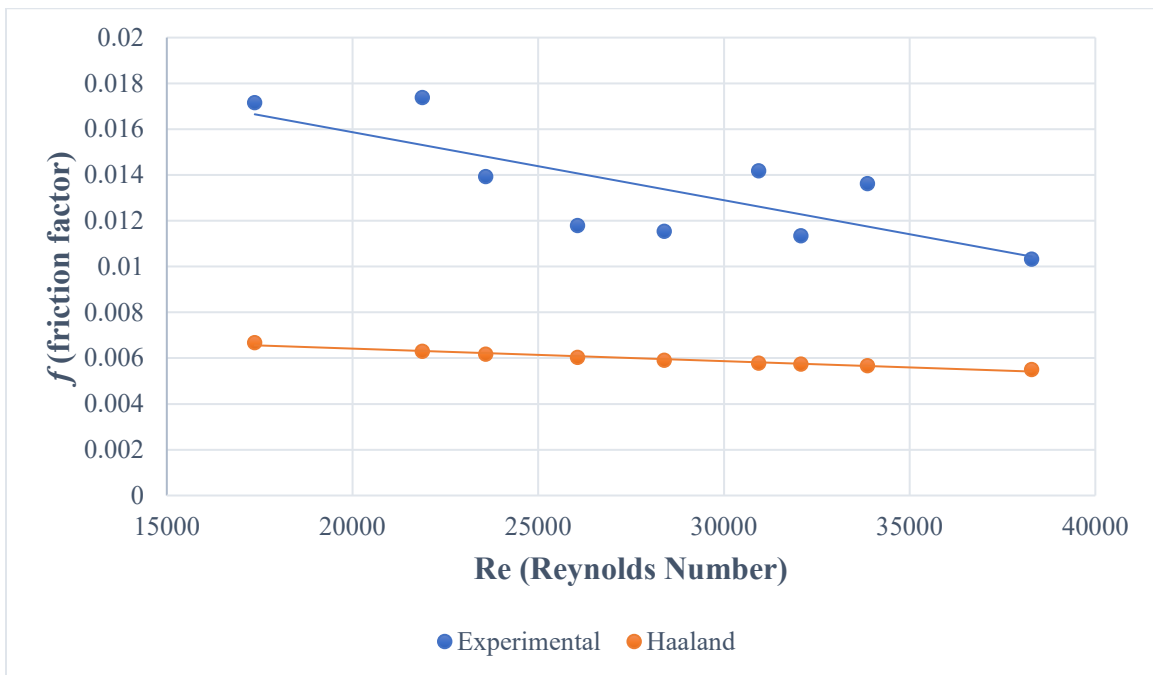


Figure 4.5: Friction factor, f , vs. Reynold's Number, Re , for Plastic Pipe.

Pipe Fitting Head Loss

In this part, we had to account for five different pipe fittings: a 180° U-bend, a 90° short elbow, a 90° square elbow, a contraction fitting, and an expansion fitting. For each fitting, three different flowrates were used to obtain the loss coefficient, K , as a function of Reynold's number, Re . Raw data can be found in Appendix I, Tables XI-XV. A graph of the results is displayed below in Figure 4.6.

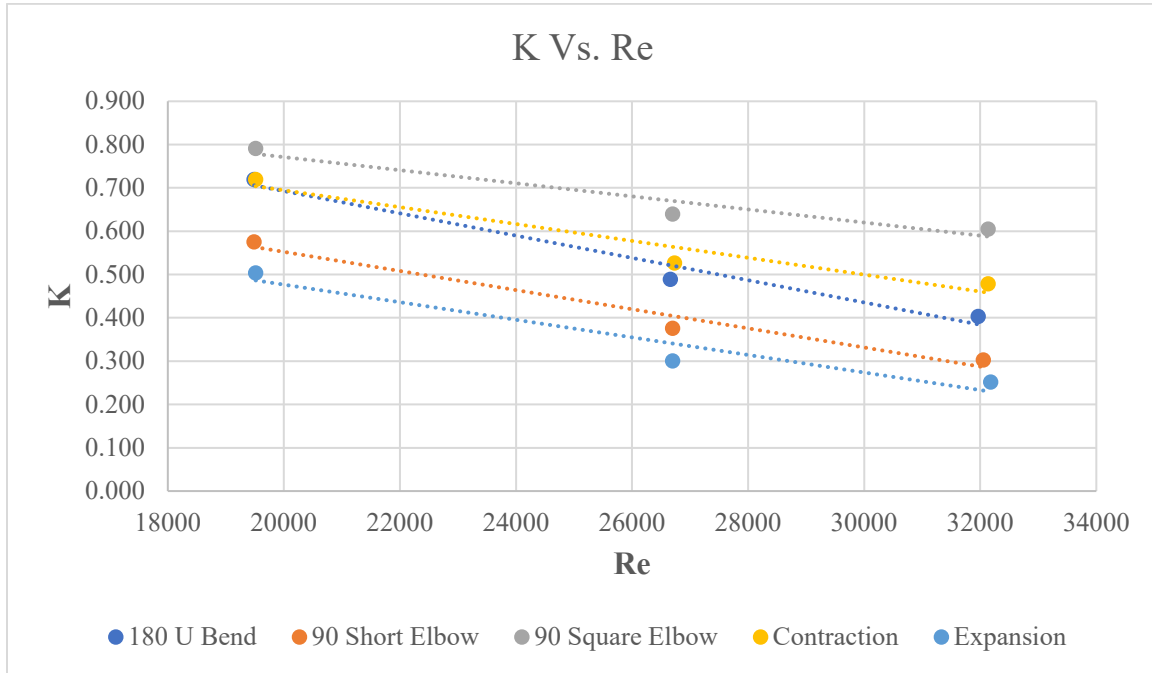


Figure 4.6: Loss Coefficient, K , of Pipe Fittings vs. Reynold's Number, Re .

5. Discussion (15 pts)

The rotameter calibration was successful, with a well-fitting linear curve obtained, as shown by Figure 4.1 and the R^2 value of 0.98 calculated in Excel. Although the curve was not perfectly linear, that can be accounted for by human error. We aimed to measure about 900 mL for each timed measurement at varying flowrates. However, in some instances, such as in higher flowrates, the water flowed out too fast and it was hard to tell when to stop the timing. Also, any small difference in timing, such as when the stopwatch was stopped versus when one person stopped collecting water, was shown by the non-linearities in Figure 4.1.

As mentioned in the Theory section, on page 3, we expected to obtain C_f values between 0.6 and 0.9 [1]. As shown in Figure 4.2, all values of C_f were between 0.52 and 0.65. These are consistent with literature values [1]. We also show in Figure 4.2 that C_f does change with varying flowrates. We would not expect this value to be constant based on equation 6. The calculation for C_f depends on both flowrate and pressure drop. If both factors change, which they do in all cases, then C_f will also change.

To compare the experimental friction factor, f , values for the pipe friction head loss to empirical data, the Haaland equation (13) was used to calculate f . To check that the f values of the Haaland equation, the Moody diagram [15] was used. Haaland equation f values corresponded almost exactly with Moody diagram (data not shown). In both empirical and experimental calculations, the friction factor shows a negative correlation with the Reynolds number, as shown in graphs 4.3-4.5. However, our experimental values for f did not correspond to the empirical data, as shown in Figures 4.3-4.5. This could be explained by the leaking from the apparatus, or possibly due to non-linearities in the calibration curve. In our f calculations, we obtained the Fanning friction factor. There is another friction factor, the Darcy friction factor, f_D . The difference between these is that f_D is four times the amount of f [2]. Because the Reynold's numbers in each data point was greater than 2900, the study was performed in turbulent flow. However, this study cannot be performed for laminar flow. Since laminar flow is independent of the roughness of the pipe surface and only due to viscous forces, we can reduce the equation for the friction factor to $f = \frac{64}{Re}$ [1].

At each flowrate and pipe fitting, we determined K, the loss coefficient, as a function of Reynold's number, Re. To determine K, we needed to find the equivalent length of diameter, L, for the fittings. We based this calculation on equation 5.7 in McCabe, which can be found on page 126. We used 50 pipe diameters, as used in ordinary practice, for this calculation [5]. The L value came to be 1.33 meters. However, there is no one value of K, as shown in Figure 4.6. For example, in the 180° U-bend, the K value at $Re = 31966$ is 0.0087, whereas at $Re = 19487$, it is 0.1439. All pipe fittings showed a trend of decreasing K value as Re increased. We believe this has to do with the pressure drop dependence of f , outlined in equation 9. According to equation 12, K is dependent only on f since L and D are both independent variables. As Re increased, the measured pressure drop decreased, therefore decreasing f . Thus, K is also decreased.

6. References (10 pts)

- [1] *Fluid Flow in Pipe Systems* - Che 381: Unit Operations Laboratory Manual
- [2] *Pressure Loss in Pipes*. Neutrium, April 29th, 2012. Accessed November 7th, 2019. https://neutrium.net/fluid_flow/pressure-loss-in-pipe/.
- [3] W. McCabe, J. Smith, and P. Harriot, *Unit Operations of Chemical Engineering*, Chapter 5 pp. 115-128, 7th ed. McGraw-Hill, 2005.
- [4] Christie John Geankoplis, Allen Hersel, Daniel Lepek. *Transport Processes and Separation Process Principles*, 5th ed. Prentice Hall, 2018.
- [5] *Fundamentals of Fluid Mechanics*. Accessed November 6th, 2019. <http://cau.ac.kr/~jjang14/FME/Chap8.pdf>
- [6] R. Byron Bird, Warren Stewart, Edwin Lightfoot. *Transport Phenomena*, 2nd ed. Wiley, 2002.

7.0 Appendix I: Data Tabulation/Graphs (10 pts)

Table I – Calibration data for the actual volumetric flow rate of water and the values read from the rotameter.

Time (min)	Volume (L)	Volumetric Flow (LPM)	SCFM	Theoretical Air (LPM)	ratio	Theoretical (LPM)	Volumetric Flow
0.066166667	0.538	8.13	13.5	11.41425	0.712354	11.41425	8.13
0.107333333	0.918	8.55	13.5	11.41425	0.749309	11.41425	8.55
0.074	0.815	11.01	16	13.528	0.814127	13.528	11.01
0.075	0.902	12.03	18	15.219	0.79024	15.219	12.03
0.059833333	0.873	14.59	20	16.91	0.862834	16.91	14.59

Table II – Calculated values of the friction factor from experimental results, friction factor from the Haaland equation, and the head loss through the copper ½ inch pipe.

Copper tubing		size (in) = 0.5		k _s (m) = 1.5E-06					
Din (in) = 0.402		Din (m) = 0.010		L (m) = 1.22					
Flowrate (LPM)	Actual Flowrate (m ³ /s)	delta P (psi)	Viscosity (Pa*s)	Re	f (exp)	f (Haaland)	h _L (m ² /s ²)	Temperature (°F)	
18.601	0.000268	2.1	0.0009792	34076	0.00566	0.00590	3.63	69.6	
16.91	0.000239	1.72	0.0009586	30963	0.00586	0.00602	2.97	71.2	
12.6825	0.000165	1.01	0.0009509	21524	0.00723	0.00651	1.75	71.8	

Table III – Calculated values of the friction factor from experimental results, friction factor from the Haaland equation, and the head loss through the copper $\frac{3}{4}$ inch pipe.

Copper tubing		size (in) = 0.75		k _s (m) = 1.5E-06				
Din (in) = 0.666		Din (m) = 0.017		L (m)= 1.22				
Flowrate (LPM)	Actual Flowrate (m ³ /s)	delta P (psi)	Viscosity (Pa*s)	Re	f (exp)	f (Haaland)	h _L (m ² /s ²)	Temperature (°F)
19.4465	0.000283	0.69	0.0009753	21790	0.02083	0.00642	1.19	69.9
17.7555	0.000254	0.58	0.0009573	19876	0.02185	0.00656	1.00	71.3
12.6825	0.000165	0.35	0.0009483	13027	0.03127	0.00727	0.61	72

Table IV – Calculated values of the friction factor from experimental results, friction factor from the Haaland equation, and the head loss through the copper 1 inch pipe.

Copper tubing		size (in) = 1.125		k _s (m) = 1.5E-06				
Din (in) = 1.025		Din (m) = 0.026		L (m)= 1.22				
Flowrate (LPM)	Actual Flowrate (m³/s)	delta P (psi)	Viscosity (Pa*s)	Re	f (exp)	f (Haaland)	h _L (m²/s²)	Temperature (°F)
40.584	0.000654	0.18	0.0009740	32727	0.00881	0.00581	0.31	70
33.82	0.000535	0.14	0.0009561	27295	0.01023	0.00605	0.24	71.4
25.365	0.000387	0.1	0.0009471	19923	0.01397	0.00651	0.17	72.1

Table V – Calculated values of the friction factor from experimental results, friction factor from the Haaland equation, and the head loss through the plastic $\frac{3}{8}$ inch pipe.

Plastic Pipe		size (in) = 0.375		k _s (m) = 0.0E+00				
Din (in) = 0.493		Din (m) = 0.013		L (m)= 1.22				
Flowrate (LPM)	Actual Flowrate (m ³ /s)	delta P (psi)	Viscosity (Pa*s)	Re	f (exp)	f (Haaland)	h _L (m ² /s ²)	Temperature (°F)
18.601	0.000268	2.72	0.0009818	27714	0.02034	0.00594	4.70	69.4
16.91	0.000239	2.26	0.0009599	25214	0.02136	0.00608	3.91	71.1
12.6825	0.000165	1.36	0.0009509	17551	0.02703	0.00665	2.35	71.8

Table VI – Calculated values of the friction factor from experimental results, friction factor from the Haaland equation, and the head loss through the plastic $\frac{1}{2}$ inch pipe.

Plastic Pipe		size (in) = 0.5		k _s (m) = 0.0E+00				
Din (in) = 0.622		Din (m) = 0.016		L (m)= 1.22				
Flowrate (LPM)	Actual Flowrate (m³/s)	delta P (psi)	Viscosity (Pa*s)	Re	f (exp)	f (Haaland)	h _L (m²/s²)	Temperature (°F)
19.4465	0.000283	0.8	0.0009779	23270	0.01717	0.00620	1.38	69.7
17.7555	0.000254	0.66	0.0009573	21282	0.01768	0.00634	1.14	71.3
12.6825	0.000165	0.42	0.0009496	13930	0.02668	0.00706	0.73	71.9

Table VII – Calculated values of the friction factor from experimental results, friction factor from the Haaland equation, and the head loss through the plastic 1 inch pipe.

Plastic Pipe		size (in) = 1		k_s (m) = 0.0E+00				
Din (in) = 1.049		Din (m) = 0.027		L (m)= 1.22				
Flowrate (LPM)	Actual Flowrate (m ³ /s)	delta P (psi)	Viscosity (Pa*s)	Re	f (exp)	f (Haaland)	h_L (m ² /s ²)	Temperature (°F)
40.584	0.000654	0.27	0.0009753	31936	0.01484	0.00574	0.47	69.9
33.82	0.000535	0.2	0.0009561	26671	0.01640	0.00600	0.35	71.4
25.365	0.000387	0.13	0.0009458	19494	0.02039	0.00648	0.22	72.2

Table VIII – Calculated values of the friction factor from experimental results, friction factor from the Haaland equation, and the head loss through the stainless steel 3/8 inch pipe.

SS Pipe		size (in) = 0.375		k_s (m) = 4.6E-05				
Din (in) = 0.493		Din (m) = 0.013		L (m)= 1.22				
Flowrate (LPM)	Actual Flowrate (m ³ /s)	delta P (psi)	Viscosity (Pa*s)	Re	f (exp)	f (Haaland)	h_L (m ² /s ²)	Temperature (°F)
18.601	0.000268	1.15	0.0009792	27786	0.00860	0.00913	1.99	69.6
16.91	0.000239	0.78	0.0009586	25247	0.00737	0.00919	1.35	71.2
12.6825	0.000165	0.53	0.0009509	17551	0.01053	0.00944	0.92	71.8

Table IX – Calculated values of the friction factor from experimental results, friction factor from the Haaland equation, and the head loss through the stainless steel ½ inch pipe.

SS Pipe		size (in) = 0.5		k_s (m) = 4.6E-05				
Din (in) = 0.622		Din (m) = 0.016		L (m)= 1.22				
Flowrate (LPM)	Actual Flowrate (m ³ /s)	delta P (psi)	Viscosity (Pa*s)	Re	f (exp)	f (Haaland)	h_L (m ² /s ²)	Temperature (°F)
19.4465	0.000283	0.51	0.0009753	23332	0.01095	0.00877	0.88	69.9
17.7555	0.000254	0.43	0.0009573	21282	0.01152	0.00884	0.74	71.3
12.6825	0.000165	0.26	0.0009471	13968	0.01652	0.00922	0.45	72.1

Table X – Calculated values of the friction factor from experimental results, friction factor from the Haaland equation, and the head loss through the stainless steel 1 inch pipe.

SS Pipe		size (in) = 1		k_s (m) = 4.6E-05				
Din (in) = 1.049		Din (m) = 0.027		L (m)= 1.22				
Flowrate (LPM)	Actual Flowrate (m ³ /s)	delta P (psi)	Viscosity (Pa*s)	Re	f (exp)	f (Haaland)	h_L (m ² /s ²)	Temperature (°F)
40.584	0.000654	0.12	0.0009740	31978	0.00660	0.00770	0.21	70
33.82	0.000535	0.1	0.0009561	26671	0.00820	0.00783	0.17	71.4
25.365	0.000387	0.08	0.0009458	19494	0.01255	0.00810	0.14	72.2

Table XI – Calculated values of the friction factor from experimental results, the loss coefficient (K), and the head loss through a 1 inch 180° U-bend.

180° U-Bend		size (in) =		k_s (m) = 1.5E-06					
Din (in) =		Din (m) = 0.0267		L (m)= 1.33					
Flowrate (SCFM)	Actual Flowrate (m ³ /s)	delta P (Pa)	Viscosity (Pa*s)	Re	v (m/s)	f (exp)	K	h_L (m ² /s ²)	Temperature (°F)
40.584	0.000654	1098.373984	0.0009728	31966	1.16857	0.00807	0.403	0.275	70.1
33.82	0.000535	892.4288618	0.0009548	26661	0.95661	0.00978	0.489	0.224	71.5
25.365	0.000387	686.4837398	0.0009445	19487	0.69167	0.01439	0.720	0.172	72.3

Table XII – Calculated values of the friction factor from experimental results, the loss coefficient (K), and the head loss through a 1 inch 90° short radius elbow.

90° Short Radius Elbow		size (in) =		k_s (m) = 1.5E-06					
Din (in) =		Din (m) = 0.0267		L (m)= 1.33					
Flowrate (SCFM)	Actual Flowrate (m ³ /s)	delta P (Pa)	Viscosity (Pa*s)	Re	v (m/s)	f (exp)	K	h_L (m ² /s ²)	Temperature (°F)
40.584	0.000654	823.7804878	0.0009702	32051	1.16857	0.00605	0.303	0.207	70.3
33.82	0.000535	686.4837398	0.0009535	26697	0.95661	0.00752	0.376	0.172	71.6
25.365	0.000387	549.1869919	0.0009445	19487	0.69167	0.01151	0.576	0.138	72.3

Table XIII – Calculated values of the friction factor from experimental results, the loss coefficient (K), and the head loss through a 1 inch 90° square elbow.

90° Square Elbow		size (in) =		k_s (m) = 1.5E-06					
Din (in) =		Din (m) = 0.0267		L (m)= 1.33					
Flowrate (SCFM)	Actual Flowrate (m ³ /s)	delta P (Pa)	Viscosity (Pa*s)	Re	v (m/s)	f (exp)	K	h_L (m ² /s ²)	Temperature (°F)
40.584	0.000654	1647.560976	0.0009676	32136	1.16857	0.01210	0.605	0.413	70.5
33.82	0.000535	1167.022358	0.0009535	26697	0.95661	0.01279	0.640	0.293	71.6
25.365	0.000387	755.1321138	0.0009432	19514	0.69167	0.01583	0.792	0.189	72.4

Table XIV – Calculated values of the friction factor from experimental results, the loss coefficient (K), and the head loss through a contraction.

Contraction		size (in) =		k_s (m) = 1.5E-06					
Din (in) =		Din (m) = 0.0267		L (m)= 1.33					
Flowrate (SCFM)	Actual Flowrate (m ³ /s)	delta P (Pa)	Viscosity (Pa*s)	Re	v (m/s)	f (exp)	K	h_L (m ² /s ²)	Temperature (°F)
40.584	0.000654	1304.319106	0.0009676	32136	1.16857	0.00958	0.479	0.327	70.5
33.82	0.000535	961.0772358	0.0009522	26733	0.95661	0.01053	0.527	0.241	71.7
25.365	0.000387	686.4837398	0.0009432	19514	0.69167	0.01439	0.720	0.172	72.4

Table XV – Calculated values of the friction factor from experimental results, the loss coefficient (K), and the head loss through an expansion.

Expansion		size (in) =		k_s (m) = 1.5E-06					
Din (in) =		Din (m) = 0.0267		L (m)= 1.33					
Flowrate (SCFM)	Actual Flowrate (m ³ /s)	delta P (Pa)	Viscosity (Pa*s)	Re	v (m/s)	f (exp)	K	h_L (m ² /s ²)	Temperature (°F)
40.584	0.000654	-686.4837398	0.0009663	32179	1.16857	0.00504	0.252	0.172	70.6
33.82	0.000535	-549.1869919	0.0009535	26697	0.95661	0.00602	0.301	0.138	71.6
25.365	0.000387	-480.5386179	0.0009432	19514	0.69167	0.01007	0.504	0.120	72.4

Table XVI – Calculated values of the orifice coefficient through a 0.478 in orifice meter.

Orifice meter		size (in) = 0.5		k_s (m) = 2.1E-07			
Din (in) = 0.478		Din (m) = 0.0121		L (m)=			
Flowrate (SCFM)	Actual Flowrate (m ³ /s)	delta P (Pa)	Viscosity (Pa*s)	Re	A0	Cf	Temperature (°F)
40.584	0.000654	37756.60569	0.0009702	70697	0.00011	0.65	70.3
33.82	0.000535	26086.38211	0.0009535	58889	0.00011	0.64	71.6
12.6825	0.000165	3775.660569	0.0009509	18163	0.00011	0.52	71.8
13.528	0.000179	4187.550813	0.0009483	19852	0.00011	0.54	72.0
25.365	0.000387	14347.51016	0.0009432	43043	0.00011	0.63	72.4

8.0. Appendix II: Error Analysis (10 pts)

For the calibration of the rotameter, the flow rate through the pipes very fast and made it difficult to accurately measure the time it took to fill a certain volume. This creates some uncertainty in the actual flow rate through the system at different rotameter values. We did not get enough data points to do a standard deviation, but we will assume that there is a $\pm 10\%$ on the actual flow rate of water through the pipes.

When some of the pressure drop measurements were being made, the pressure taps were leaking, indicating a slight loss of pressure due to the escaping fluid. This can affect the reading of the pressure drop and will cause error in the calculation for head loss for the pipes and fittings that had the leaking pressure taps.

The error between the experimental values of the friction factor for the pipes and the friction factor as calculated by the Haaland equation are tabulated below:

Table XVII – Calculated errors between the experimental friction factors for the pipe and the predicted friction factors using the Haaland equation. Error is calculated as $|f_{\text{Haaland}} - f_{\text{exp}}|/f_{\text{Haaland}} * 100 = \text{Error } \%$.

	f (exp)	f (Haaland)	Error		f (exp)	f (Haaland)	Error		f (exp)	f (Haaland)	Error
Copper	0.00566	0.00590	4%	S. Steel	0.02034	0.00594	242%	plastic	0.0086	0.00913	6%
	0.00586	0.00602	3%		0.02136	0.00608	251%		0.0074	0.00919	20%
	0.00723	0.00651	11%		0.02703	0.00665	306%		0.0105	0.00944	12%
	0.02083	0.00642	225%		0.01717	0.00620	177%		0.0109	0.00877	25%
	0.02185	0.00656	233%		0.01768	0.00634	179%		0.0115	0.00884	30%
	0.03127	0.00727	330%		0.02668	0.00706	278%		0.0165	0.00922	79%
	0.00881	0.00581	52%		0.01484	0.00574	158%		0.0066	0.00770	14%
	0.01023	0.00605	69%		0.01640	0.00600	174%		0.0082	0.00783	5%
	0.01397	0.00651	114%		0.02039	0.00648	215%		0.0126	0.00810	55%

Not enough repeat trials were performed on the data for the pipes in order to calculate a standard deviation. The errors in our experimental data range from 3% to 306% when compared to the predicted friction factors from the Haaland equation.

For the orifice meter, the standard range of orifice coefficients (C_f) range from 0.6 to 0.9. For the error analysis, we will assume the lower end of the range is the theoretical value (0.6). Looking at **Table XVI**, the following errors were tabulated:

Table XVIII – Calculated errors between the experimental orifice coefficient for the orifice meter and the predicted orifice coefficients from the theoretical values. Error is calculated as $|C_f - C_{f,exp}|/C_f * 100 = \text{Error } \%$.

	Cf (exp)	Cf	 Error
Orifice	0.65331	0.60000	9%
	0.64341	0.60000	7%
	0.52023	0.60000	13%
	0.53846	0.60000	10%
	0.62729	0.60000	5%

The error in the C_f value for the orifice is within the $\pm 15\%$ of the theoretical value.

A standard deviation will be performed on any data that has three repeat trials. Any extra data that is within the 95% confidence interval will be kept to support the conclusion of our data. The standard deviation formulation is given below:

$$\sigma = \sqrt{\frac{\sum(x - \bar{x})^2}{n}}$$

Where σ is the standard deviation, n is the sample size, and $\bar{x}$ the average of each data point taken throughout the experiments. To calculate if the collected data is within the 95% confidence interval, the following formula is used:

$$Ci = Z * n$$

Where Z is the z-value at the 95% confidence interval.

9.0 – Appendix III: Sample Calculations

Rotameter Calibration

$$LPM \text{ with conversion} = SCFM * 0.8455$$

$$LPM \text{ with conversion} = 13.5 * 0.8455$$

$$LPM \text{ with conversion} = 11.41$$

$$\text{Actual LPM} = 8.13$$

*New conversion needed so a calibration curve was made

$$LPM \text{ actual} = 1.0519 * LPM \text{ with conversion} - 3.4627$$

$$LPM \text{ actual} = 1.0519 * (22 SCFM * 0.8455) - 3.4627$$

$$LPM \text{ actual} = 16.10$$

*The following samples are for the 0.5 inch by 48 inch pipe with a flowrate of $3.28 * 10^{-4} \text{ m}^3/\text{s}$.

Reynolds number

$$Re = \frac{\rho D v}{\mu}$$

$$Re = \frac{997 * 0.01 * (\frac{2.68 * 10^{-4}}{\pi * 0.005^2})^2}{9.79 * 10^{-4}}$$

$$Re = 34,076$$

Friction factor for straight pipes using Haaland Eq [6].

$$\frac{1}{\sqrt{f}} = -3.6 \log_{10} \left\{ \frac{6.9}{Re} + \frac{k}{3.707D} \right\}$$

$$f = \left(-3.6 \log_{10} \left\{ \frac{6.9}{34,076} + \frac{1.5 * 10^{-6}}{3.707 * 0.01} \right\} \right)^{-2}$$

$$f = 0.00590$$

Friction factor for straight pipes using experimental data

$$f = \frac{\Delta p D}{2L\rho V^2} \quad [5]$$

$$f = \frac{2.1 * 6894.76 * 0.01}{2 * 1.22 * 997 * (\frac{2.68 * 10^{-4}}{\pi * 0.005^2})^2}$$

$$f = 0.00566$$

Head loss

$$h_L = f * \frac{LV^2}{2D}$$

$$h_L = 0.00567 * \frac{1.22 * (\frac{2.68 * 10^{-4}}{\pi * 0.005^2})^2}{2 * 0.01}$$

$$h_L = 3.63 \frac{m^2}{s^2}$$

*The following sample calculations are for the 180° U-bend

Friction factor using experimental data

Same method as shown above for straight pipe

Loss coefficient K

$$K = f * \frac{L}{D}$$

$$K = 0.00807 * \frac{1.33}{0.0267}$$

$$K = 0.403$$

Head loss

$$h_L = K * \frac{V^2}{2}$$

$$h_L = 0.403 * \frac{1.17^2}{2}$$

$$h_L = 0.275 \frac{m^2}{s^2}$$

*The following calculations are for the orifice

Area of the orifice

$$A_0 = \pi * \left(\frac{0.0121}{2}\right)^2$$

$$A_0 = 0.00011 m^2$$

Flow coefficient

$$Q = C_f A_0 \sqrt{2\Delta p / \rho}$$

$$C_f = \frac{Q}{A_0 \sqrt{2\Delta p / \rho}}$$

$$C_f = \frac{6.54 * 10^{-4}}{1.15 * 10^{-4} \sqrt{2 * 37,756.6 / 997}}$$

$$C_f = 0.65$$

10. Appendix IV: Individual Team Contributions

NAME: James Fell

	TIME (HOURS)
Operator (Both Lab Days)	4 hrs
Pre-Lab Editing	
Experimental Objective (Prelab)	0.5 hrs
Apparatus (prelab)	
Materials & Supplies (prelab)	0.5 hrs
Procedure (prelab)	0.5 hrs
Project Safety Evaluation (prelab)	0.5 hrs
Final Lab Editing	
Executive Summary (final lab)	
Introduction (Final lab)	
Theory (Final Lab)	
Results (final lab)	2 hrs
Discussion (final lab)	
References (final and/or prelab)	0.5 hrs
Data Tabulation/Graphs (final)	2 hrs
Error Analysis (final lab)	0.5 hrs
Sample Calculations (final Lab)	0.5 hrs
Power Point Presentation	
Total	11.5 hrs

NAME: Jasmine Villegas

	TIME (HOURS)
Operator (Both Lab Days)	4 hrs
Pre-Lab Editing	0.5 hrs
Experimental Objective (Prelab)	
Apparatus (prelab)	
Materials & Supplies (prelab)	
Procedure (prelab)	1.5 hrs
Project Safety Evaluation (prelab)	

Final Lab Editing	1 hrs
Executive Summary (final lab)	
Introduction (Final lab)	
Theory (Final Lab)	
Results (final lab)	3 hrs
Discussion (final lab)	
References (final and/or prelab)	1 hrs
Data Tabulation/Graphs (final)	
Error Analysis (final lab)	
Sample Calculations (final Lab)	
Power Point Presentation	
Total	10 hrs

NAME: Mariana St. Peter

	TIME (HOURS)
Operator (Both Lab Days)	4 hrs
Pre-Lab Editing	
Experimental Objective (Prelab)	
Apparatus (prelab)	
Materials & Supplies (prelab)	
Procedure (prelab)	0.5 hrs
Project Safety Evaluation (prelab)	
Final Lab Editing	2 hrs
Executive Summary (final lab)	2 hrs
Introduction (Final lab)	
Theory (Final Lab)	
Results (final lab)	
Discussion (final lab)	2.5
References (final and/or prelab)	
Data Tabulation/Graphs (final)	
Error Analysis (final lab)	
Sample Calculations (final Lab)	
Power Point Presentation	

Total	11.5 hrs
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NAME: William Rawson

	TIME (HOURS)
Operator (Both Lab Days)	4 hrs
Pre-Lab Editing	
Experimental Objective (Prelab)	0.5 hrs
Apparatus (prelab)	1 hrs
Materials & Supplies (prelab)	0.5 hrs
Procedure (prelab)	0.5 hrs
Project Safety Evaluation (prelab)	0.5 hrs
Final Lab Editing	
Executive Summary (final lab)	
Introduction (Final lab)	2
Theory (Final Lab)	2.5
Results (final lab)	
Discussion (final lab)	
References (final and/or prelab)	
Data Tabulation/Graphs (final)	
Error Analysis (final lab)	
Sample Calculations (final Lab)	
Power Point Presentation	
Total	11.5 hrs